Dong-Geun Kim

Ph.D. Candidate | POSTECH Interaction Lab

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Google Scholar · Publication data checked 2026-10-19

Research interests

Tactile communication · Vibrotactile perception · Mid-air haptics

Education

Pohang University of Science and Technology (POSTECH)

Integrated M.S./Ph.D. in IT Convergence Engineering

2020 - Present

Ulsan National Institute of Science and Technology (UNIST)

B.S. in Mechanical Engineering

2014 - 2019

Research experience

Research Assistant · POSTECH Kim Lab

Sep. 2019 - May 2020

Developed high-frequency ultrasound transducers for tactile applications.

Awards

2026 · Best Paper Award Finalist
IEEE Haptics Symposium

2024 · Best ToH Short Paper Award (Finalist)
IEEE Haptics Symposium

2022 · Best Paper Award
KSC (KIISE)

Publications

Titles and years follow the Google Scholar profile where indexed; accepted papers not yet indexed are marked. Complete author names and existing PDF links are retained where available.

1. Effects of Spatiotemporal Parameters on Forearm Vibrotactile Stimulus Identification

Dong-Geun Kim, Geunho Lee, Suheon Nam, Chaeyong Park, Seungmoon Choi

2026 IEEE Haptics Symposium (HAPTICS), 1-13 · 2026

Best Paper Award Finalist

2. Investigating Luxuriousness of Steering Wheel Vibration Warning

K Kwon, SH Han, S Choi, B Jeong, J Kim, J Lee, DG Kim, W Lee

International Journal of Human-Computer Interaction, 1-21 · 2026

3. Synchronization-Enabled Perceptual Control in Wearable Vibrotactile Systems

Taeji Lee†, Dong-Geun Kim†, Hee-Sup Shin†, Myung-Kun Chung†, Jaewon Lim, Yeonho Kim, Hyein Jung, Chan Shin, Jay Young Kwak, Min-Seung Jo, Kyoung-Ho Ha, Minsu Park, Hyoyoung Jeong, Yei Hwan Jung, J. Edward Colgate, Seungmoon Choi*, Jae-Young Yoo*, John A. Rogers*, and Jin-Tae Kim*

†Co-first authors · *Co-corresponding authors

Nature Communications · 2026 · Accepted

4. Tactile Enhancement of Mid-Air Ultrasonic Stimulation by Wrist Vibration: Perceived Intensity and Pattern Recognition

Dong-Geun Kim, Seungmoon Choi

IEEE Transactions on Haptics 18 (3), 603-614 · 2025

5. A feasibility study on tactile enhancement of mid-air ultrasonic stimulation by wrist vibration

Dong-Geun Kim and Seungmoon Choi

International Conference on Human Haptic Sensing and Touch Enabled Computer ... · 2024

6. Generating real-time, selective, and multimodal haptic effects from sound for gaming experience enhancement

Gyeore Yun, Minjae Mun, Jungeun Lee, Dong-Geun Kim, Hong Z Tan, and Seungmoon Choi

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems ... · 2023

7. Sound-to-touch crossmodal pitch matching for short sounds

Dong-Geun Kim, Jungeun Lee, Gyeore Yun, Hong Z. Tan, and Seungmoon Choi

IEEE Transactions on Haptics 17 (1), 2-7 · 2023

8. MMGrip: A Handheld Multimodal Haptic Device Combining Vibration, Impact, and Shear for Realistic Expression of Contact

Dong-Geun Kim, Jungeun Lee, and Seungmoon Choi

SIGGRAPH Asia 2022 Posters, 1-2 · 2022

9. MMGrip: 사실적인 햅틱 표현을 위한 진동, 임팩트, 전단력 기반 다중촉감 햅틱 장치

Dong-Geun Kim, Jungeun Lee and Seungmoon Choi

한국정보과학회 학술발표논문집, 1038-1040 · 2022

10. Vibration-augmented buttons: Information transmission capacity and application to interaction design

Chaeyong Park, Jeongwoo Kim, Dong-Geun Kim, Seungjae Oh, and Seungmoon Choi

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems ... · 2022

Additional presentations and posters

Preserved from the existing website; not part of the current Scholar publication list.

1. Tactile Localization in Forearm-Mounted Multi-Tactor Displays: The Effect of Tactor Number and Position

Dong-Geun Kim, Suheon Nam, Geunho Lee, Chaeyong Park and Seungmoon Choi

IEEE World Haptics Conference · 2025

2. A Feasibility Study of Tactile Enhancement of Mid-Air Ultrasonic Stimulation by Wrist Vibration

Dong-Geun Kim and Seungmoon Choi

Proceedings of Korea Haptics Conference · 2024

3. Investigating the Luxuriousness of Haptic Warning on the Steering Wheel

Jeongwoo Kim, Jungeun Lee, Dong-Geun Kim, Ki-Min Kwon, Beomyoung Chung, Junseong Park, Taeuk Kang, Sangji Moon, Wonseok Leee, Sung H. Han, and Seungmoon Choi

Spring Conference, The Acoustical Society of Korea · 2024

4. Real-Time, Selective, and Multimodal Haptic Feedback from Gameplay Sound

Gyeore Yun, Jungeun Lee, Dong-Geun Kim, Minjae Mun, Hong Z. Tan, and Seungmoon Choi

IEEE World Haptics Conference · 2023

5. 음향**-**촉감 교차 모달 스펙트럼 매칭

Dong-Geun Kim, Jungeun Lee, Gyeore Yun, Hong Z. Tan, and Seungmoon Choi

Korean Institute of Next Generation Computing · 2023

Funded projects

Tactile Korean System for the Visually and Hearing Impaired (PM)

IITP | Jan. 2025 - Present

Developing a way to deliver Korean text through a wearable vibrotactile array, for people with visual or hearing impairments.

Mid-air Haptic Rendering for Digital Twin (PM)

IITP | Jan. 2023 - Dec. 2024

Real-time mid-air haptic rendering for digital twins. Studied how wrist vibration raises the perceived intensity of weak ultrasonic stimuli and improves pattern recognition.

Investigating the Luxuriousness of Haptic Warning

Hyundai Motors | Apr. 2023 - Apr. 2024

User studies on steering-wheel vibration warnings and what makes them feel luxurious.

Performance Metric for User Experience

Google | Sep. 2021 - Jul. 2023

Metrics for user experience in digital environments, including sound-to-touch matching for short sounds.

Multimodal Haptic Tool for AR Sports

KOCCA | Jun. 2020 - Dec. 2022

MMGrip, a handheld device that combines vibration, impact, and shear to render contact realistically.

Technical skills

Programming: Python, Unity (C#), MATLAB, C/C++

Hardware & research: Arduino / Embedded, Fusion 360 (CAD), 3D Printing, Signal Processing, User Study Design

Languages

Korean (Native) · English (Fluent) · TOEIC 920 / TEPS 379